

ZVNS3-c003 -p007 purification

PAGE24-01328

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Date Started: **2024-Aug-12**

Experiment Started:

Projects: **Expression;Purification;ASAP**

Related Pages:

Referenced by:

Tags:

Expression from Nathan

Pellet taken from Nathan's -80 freezer, no expression ID yet?

Updated: ZVNS3-e007

Label on bag:

Zika Helicase NS3

11/7/24 NW

121.3g

NO CLEAVAGE

IMAC

Lysis buffer - 50 mM HEPES pH 7.4, 500 mM NaCl, 5% glycerol, 0.5mM TCEP

Wash buffer 1 - 50 mM HEPES pH 7.4, 500 mM NaCl, 5% glycerol, 0.5mM TCEP, 20mM imidazole

Elution Buffer - 50 mM HEPES pH 7.4, 500 mM NaCl, 5% glycerol, 0.5mM TCEP, 500mM imidazole

Gel Filtration Buffer (SEC) - 50 mM HEPES pH 7.4, 500 mM NaCl, 5% glycerol, 0.5mM TCEP

1. Lysis buffer supplement with 1:4000 dilution of benzonase, 1mg/mL lysozyme. Incubate for 30min at RT.
2. Sonicated on ice at 50% amplitude for a total of 7-minute sonication time (4 seconds on 12 seconds off) with thick probe.
3. Clarified lysate by centrifugation at 18,000rpm, 4°C for 1 hour. Used JLA16.250 rotor. Supernatant poured into clean beaker.

IMAC

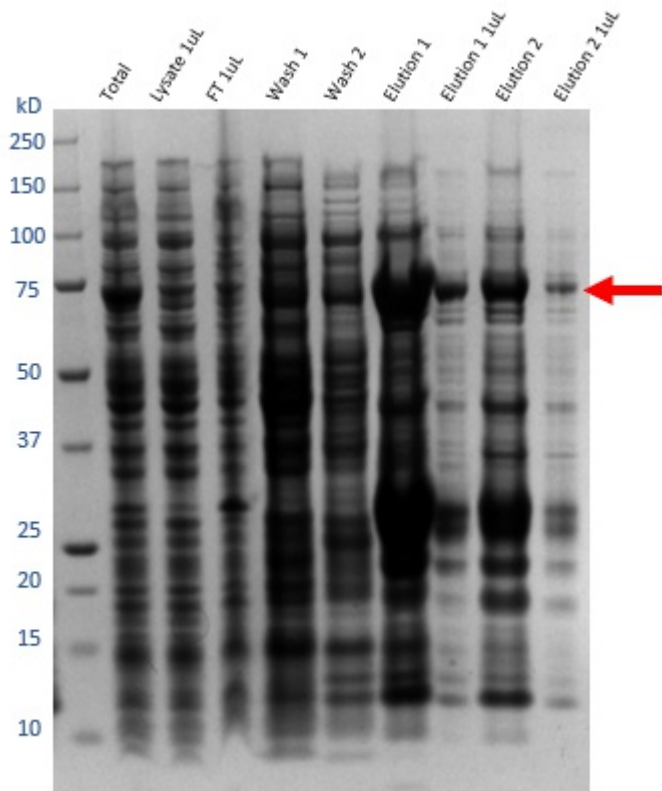
1. Wash and equilibrate 5mL bed volume of Ni Sepharose resin on gravity flow column, first with distilled water, then with wash buffer.
2. Put stopper on column. Re-suspend resin with small volumes of the lysis buffer, and pour into beaker with lysate. Stir in cold room gently for 10min.
3. Pour mixture back onto gravity flow column and let flow through.
4. Wash resin with 25mL wash buffer 1 twice.
1. Elute with 7.5mL elution buffer, 2 elutions carried out

Gel filtration

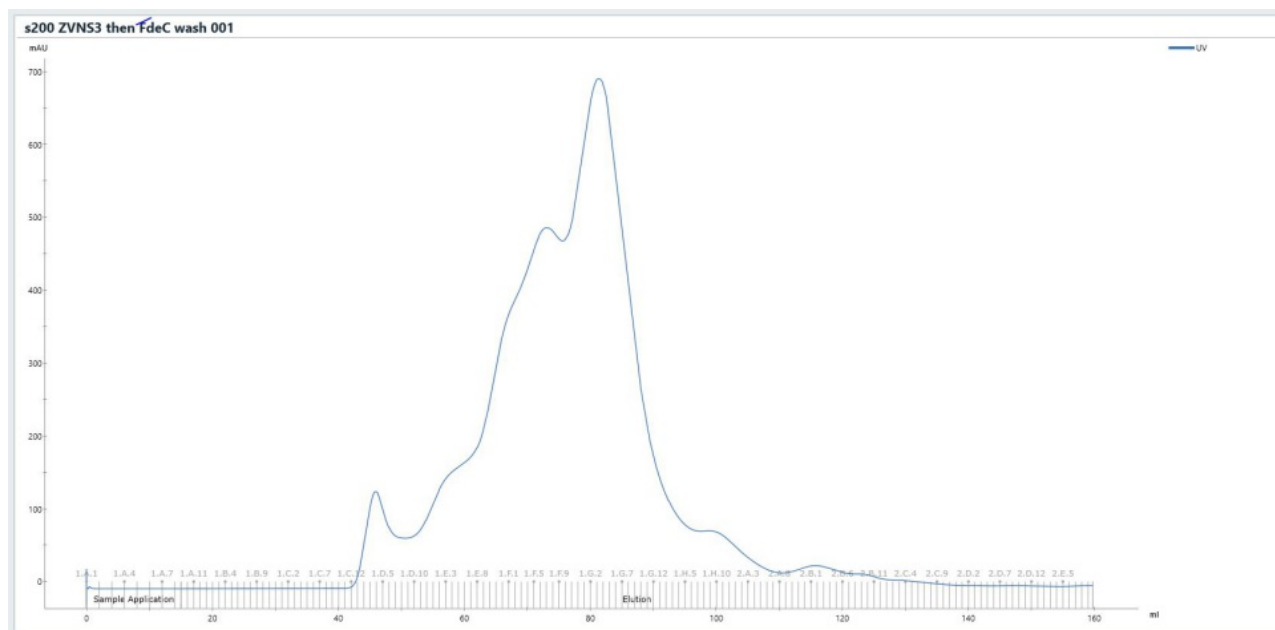
1. Elutions concentrated in 30kDa MWCO Amicon concentrators to final volume of ~1mL

2. Injected onto Superdex s200 16/60 column and run in SEC buffer at 1mL/min
3. After SEC, peaks were analysed by SDS-PAGE

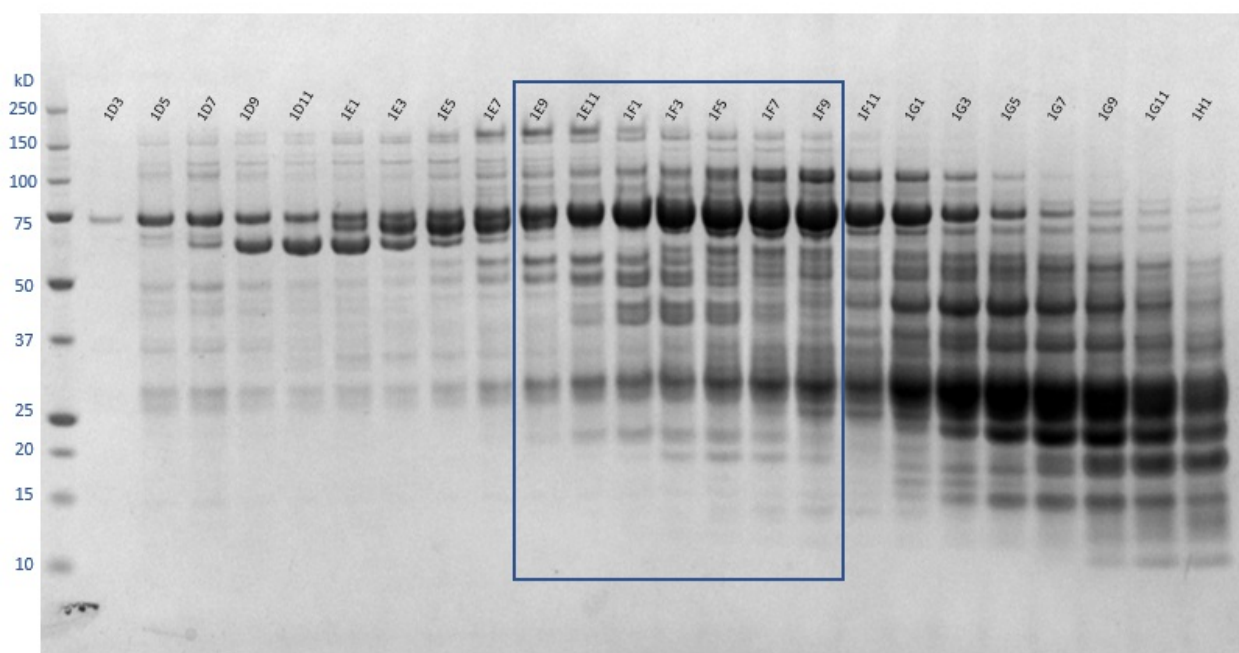
IMAC result



ZVNS3 s200



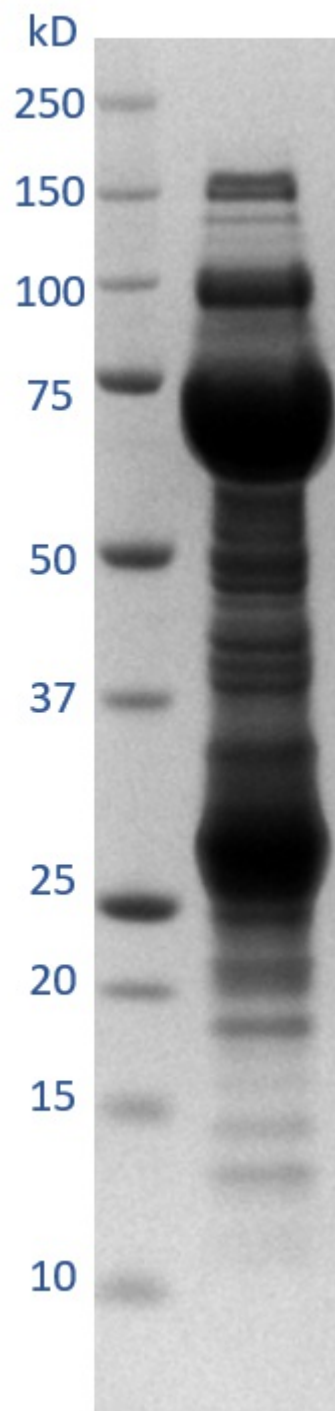
SEC fractions



Final sample

23.95 mg/mL, 12x50uL + 20uL
Total yield: 14.85mg

Final sample?



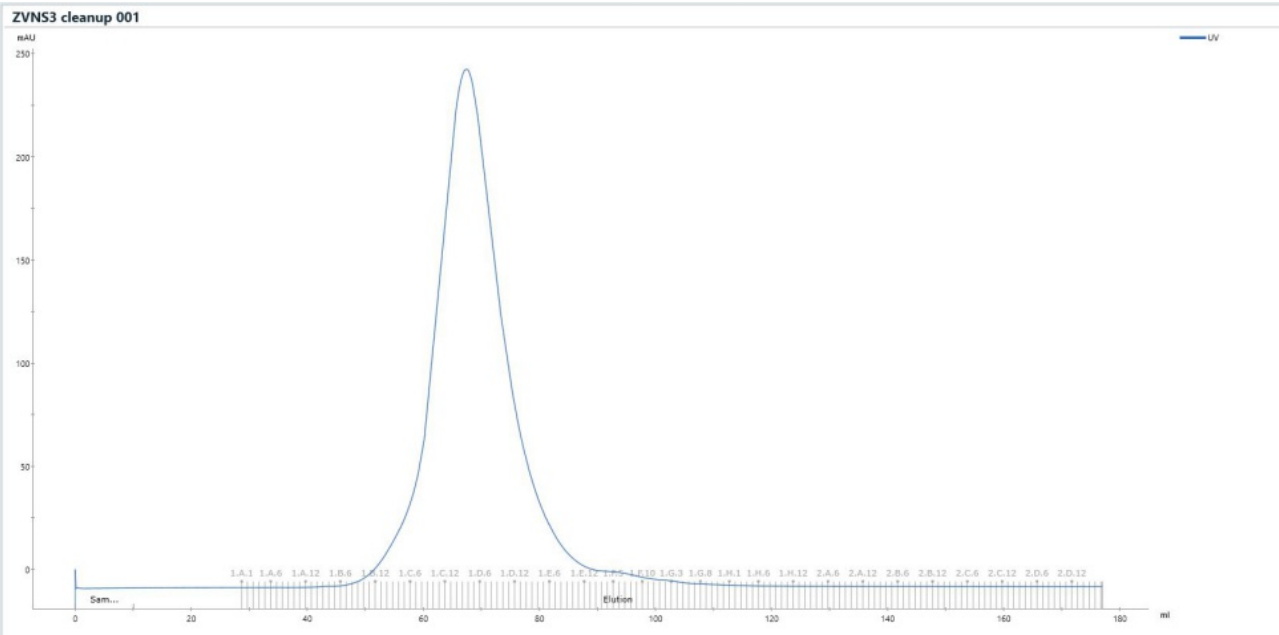
Title missing - double click to edit

Final sample looks fairly dirty. Pass down a SEPAX s300 next week to see if can be cleaned up.

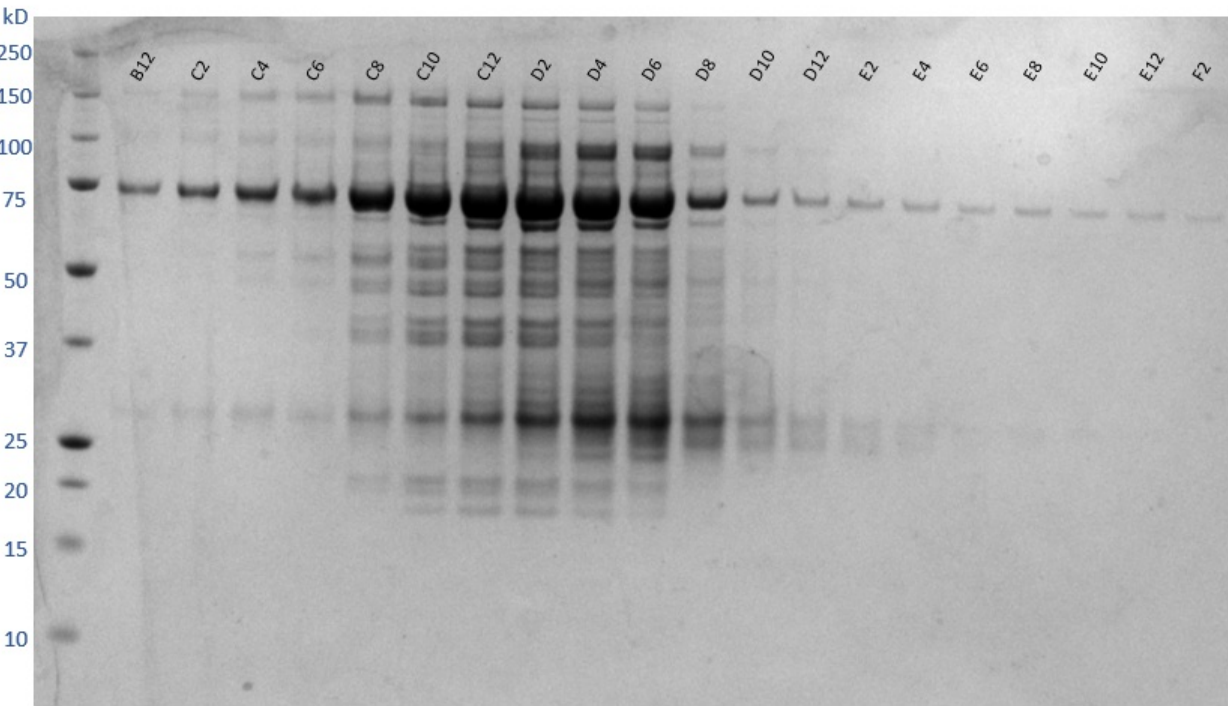
13/8/2024

All ZVNS3 samples thawed and ran down SEPAX-SRT SEC300 with guard column, at 7.5mL/min.

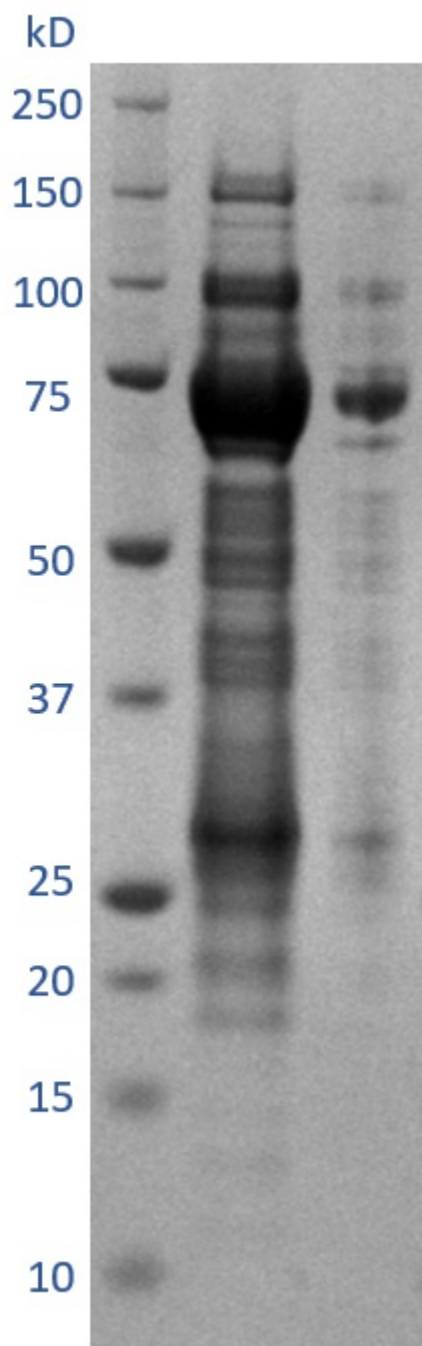
SEPAX cleanup chromatogram



SEPAX cleanup fractions



After clean-up



Final sample

Fractions B12-D8 were pooled and concentrated.

Final sample: 19.61 mg/mL, 9*50uL

stored in box 4

Was not able to separate either the higher MW or the lower MW contaminants. Possibl degradation product or things stuck to the target?